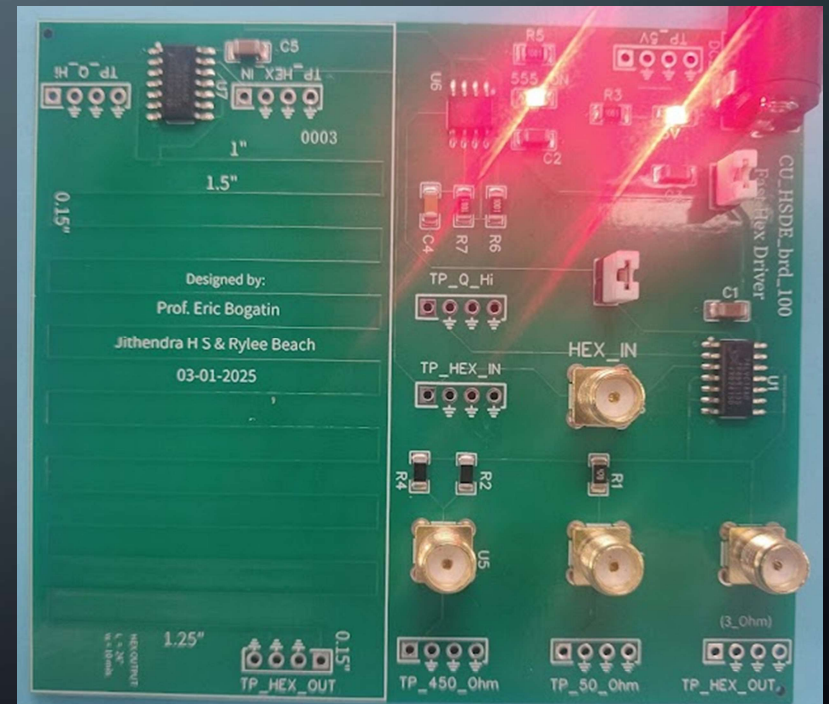
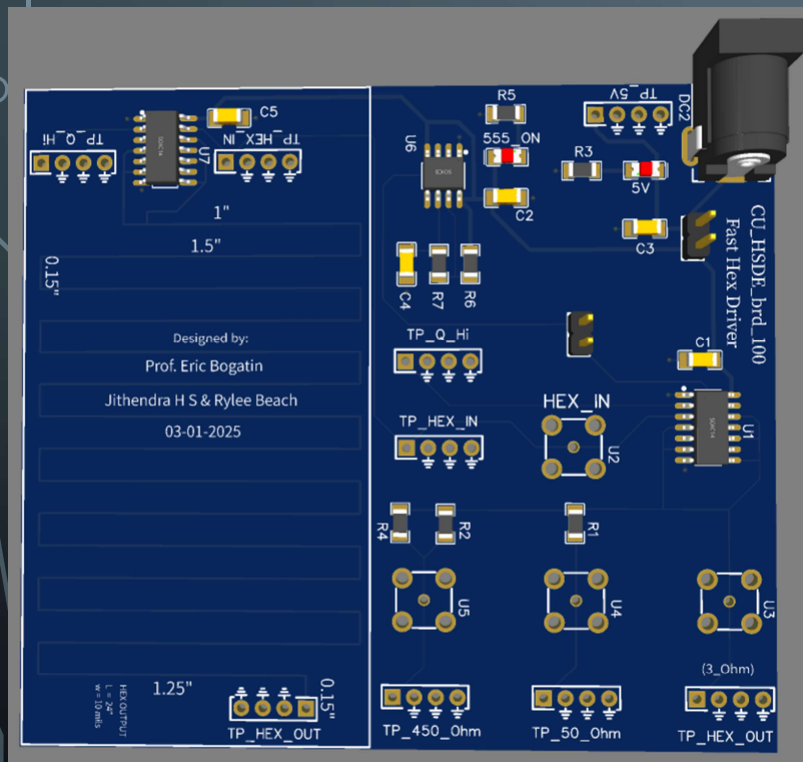


PRJ 100 : FAST HEX DRIVER





CHARACTERIZE

- 5V and 555 timer
- Q high Noise
- Hex inverter
- TD between peaks in the ringing
- 3, 50, 450ohm at 1M
- 3, 50, 450ohm at 50ohm

5V AND 555 TIMER

5V rail with no switching load

Meas
Freq(1): 531.4Hz
Pk-Pk(1): 300mV
Avg - FS(1): 5.4828V

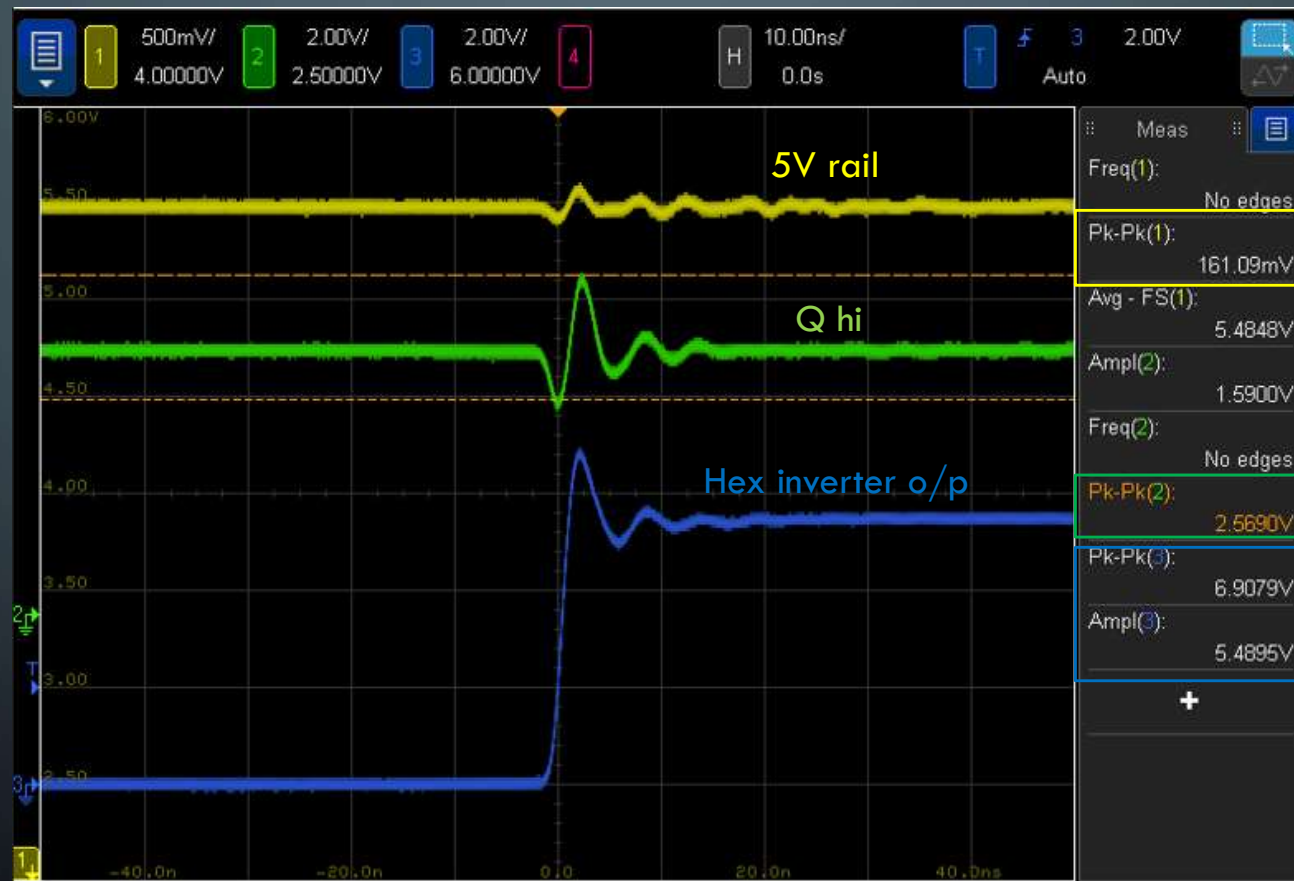
555 timer o/p

Meas
Freq(1): No signal
Pk-Pk(1): No signal
Avg - FS(1): No signal
Ampl(2): 5.49V
Freq(2): 46.638kHz

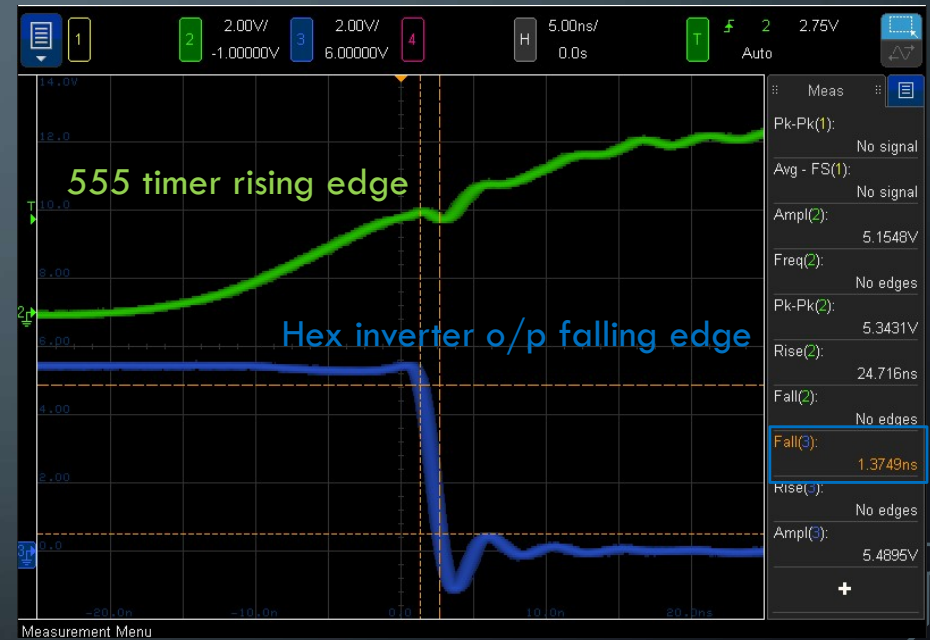
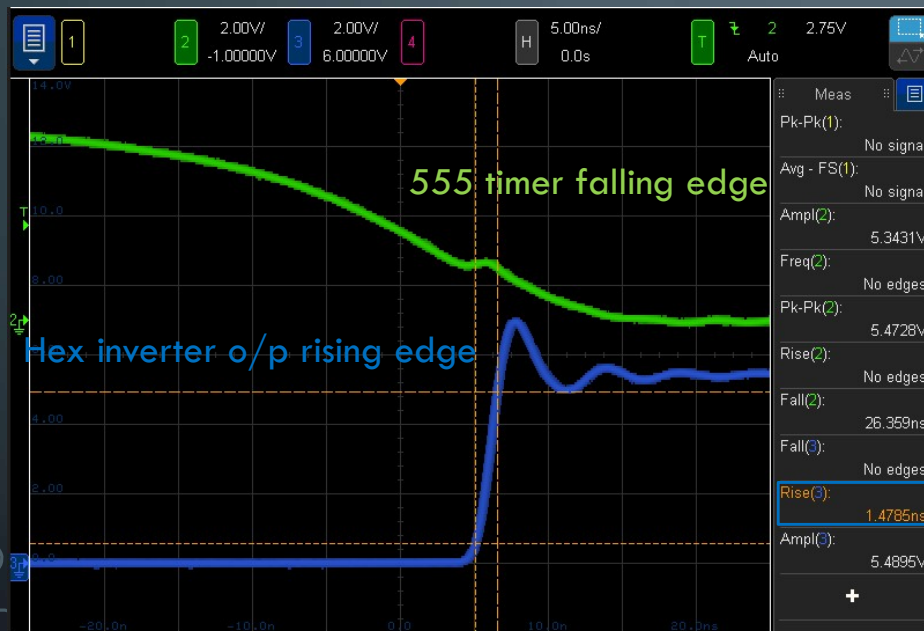
555 TIMER RISE & FALL TIME



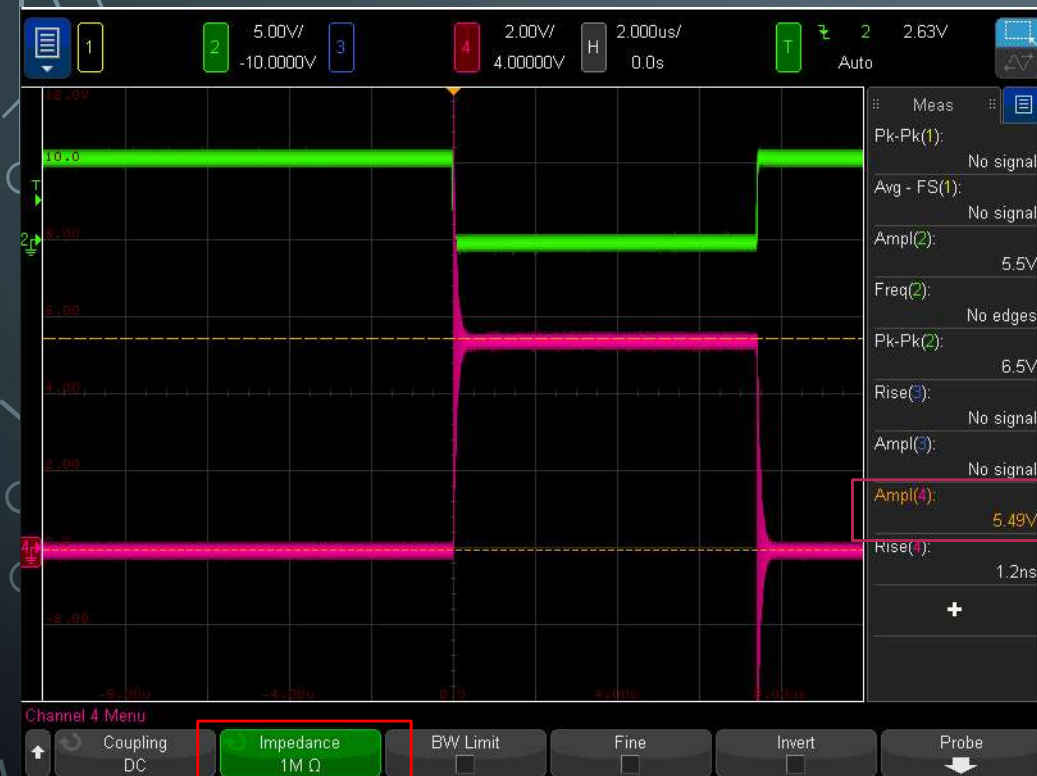
Q HI NOISE



HEX INVERTER



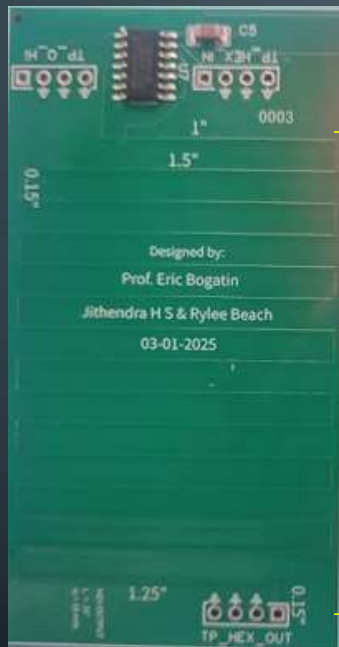
THEVENIN RESISTANCE



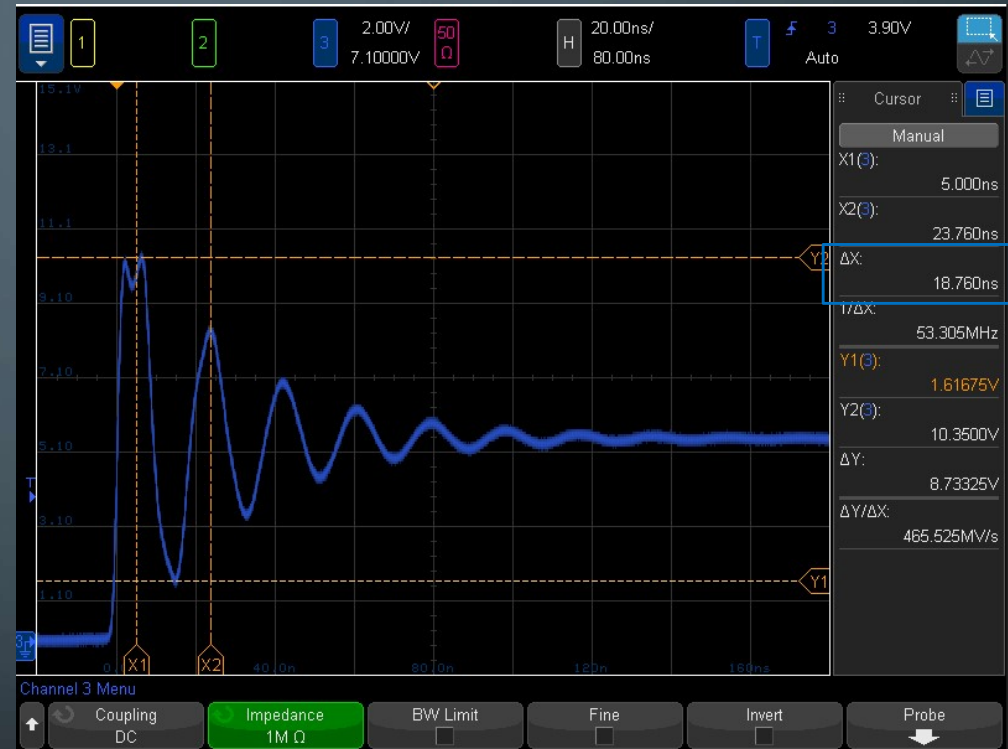
$$50 \cdot \frac{(5.49 - 5.22)}{5.22} = 2.586206897$$

→ ~3 Ω

TIME DELAY BETWEEN PEAKS IN THE RINGING

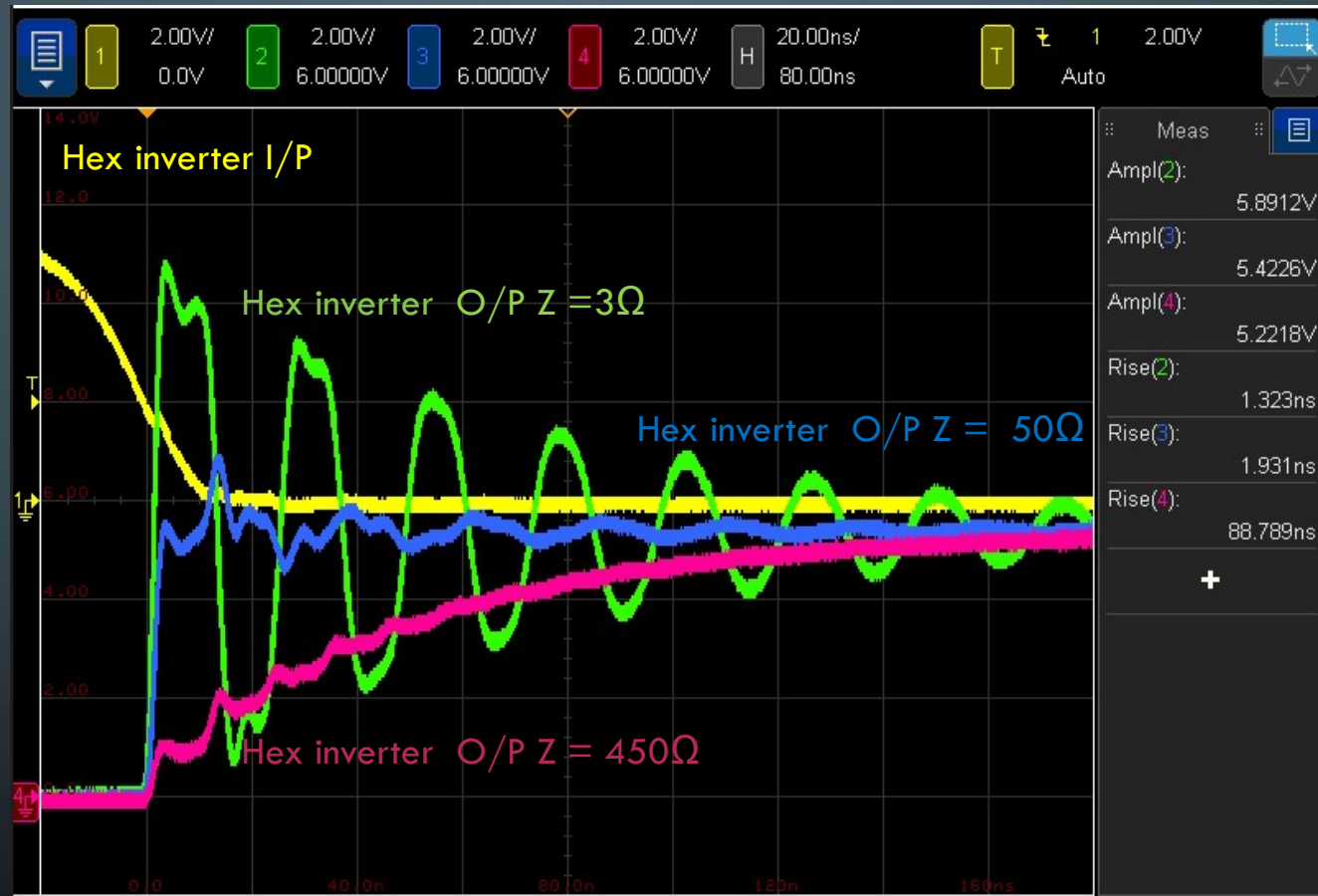


24inch long

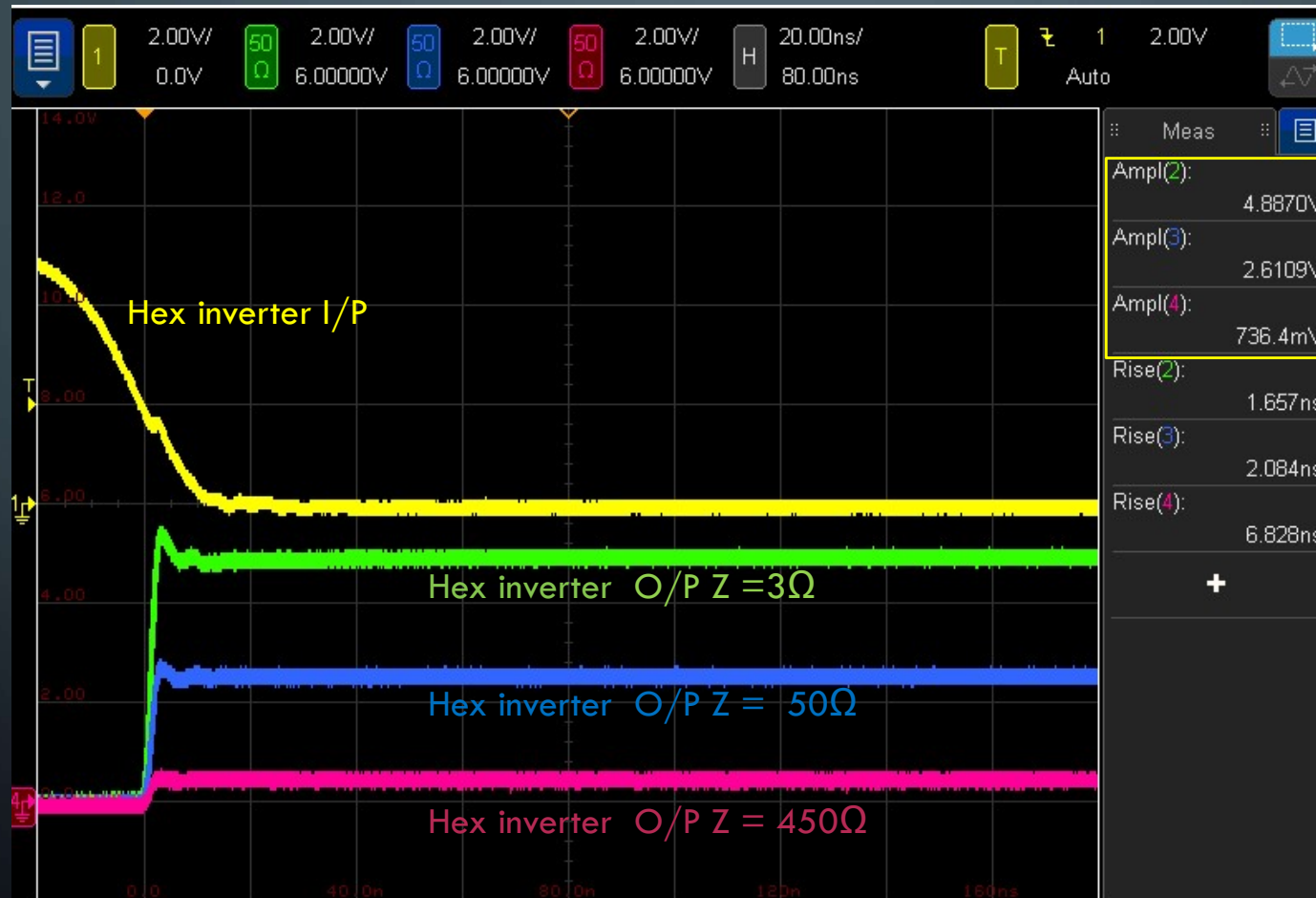


For an FR4 substrate, the signal velocity is approximately 6 inches/ns. The spatial extent of the 24-inch trace takes a time delay (TD) of 4 ns. According to the bounce diagram, peaks occur at 4TD, so the expected peak is at 16 ns. However, the measured peak was 18.76 ns due to the parasitic capacitance of the probe.

3, 50, 450 Ω OUTPUT IMPEDANCE VS 1M AT SCOPE



3, 50, 450 Ω OUTPUT IMPEDANCE VS 50 Ω AT SCOPE



Acts as a voltage divider

REFLECTION AT MATCHED 50Ω IMPEDANCE?

